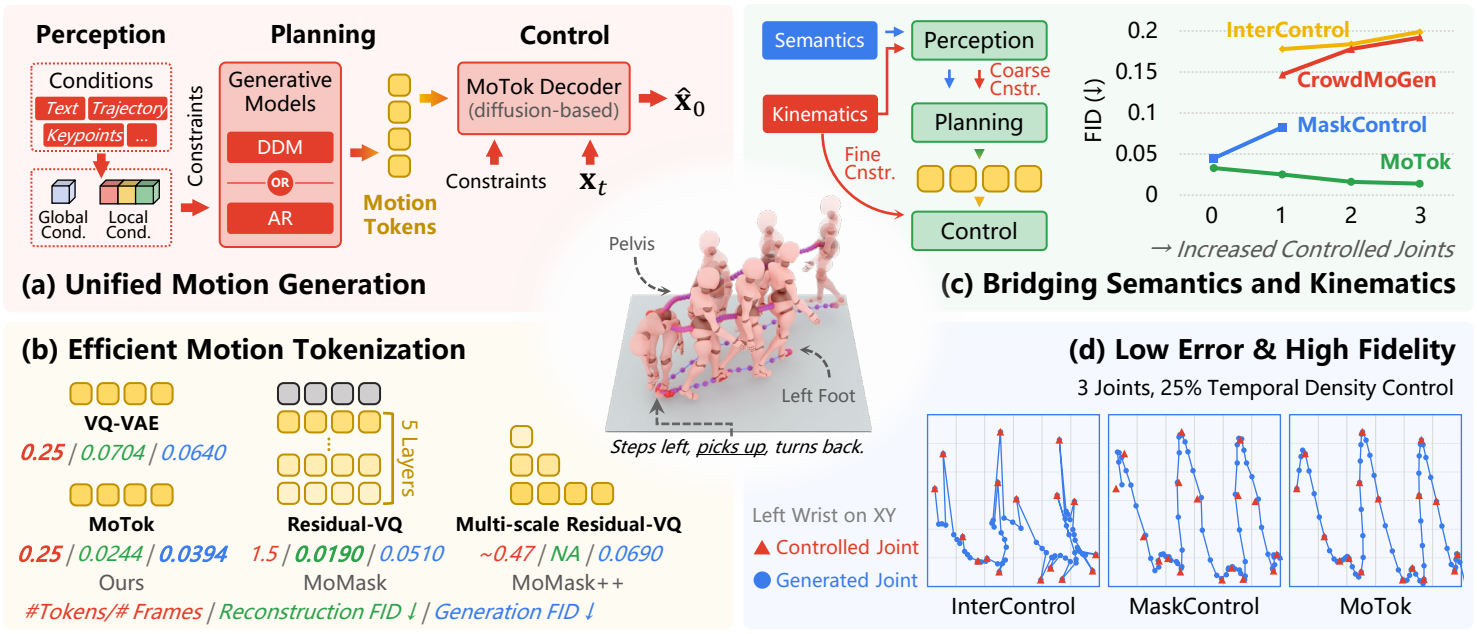
Perception
Planning
Control
Conditions
Text
Trajectory
Keypoints
...
Global Cond.
Local Cond.
Constraints
Generative Models
DDM
OR
AR
Motion Tokens
MoTok Decoder (diffusion-based)
Constraints
$\mathbf{x}_t$
$\hat{\mathbf{x}}_0$
(a) Unified Motion Generation
Semantics
Kinematics
Perception
Coarse Cnstr.
Planning
Fine Cnstr.
Control
FID (↓)
0.2
0.15
0.1
0.05
0
InterControl
CrowdMoGen
MaskControl
MoTok
0
1
2
3
→ Increased Controlled Joints
(c) Bridging Semantics and Kinematics
Pelvis
Left Foot
Steps left, picks up, turns back.
(b) Efficient Motion Tokenization
VQ-VAE
0.25 | 0.0704 | 0.0640
MoTok
0.25 | 0.0244 | 0.0394
Ours
Residual-VQ
5 Layers
1.5 | 0.0190 | 0.0510
MoMask
Multi-scale Residual-VQ
~0.47 | NA | 0.0690
MoMask++
#Tokens/# Frames | Reconstruction FID ↓ | Generation FID ↓
(d) Low Error & High Fidelity
3 Joints, 25% Temporal Density Control
Left Wrist on XY
Controlled Joint
Generated Joint
InterControl
MaskControl
MoTok